

PERFORMANCE WORK STATEMENT (PWS)

Ultrasonic Impact Treatment and Hammer Peening of HY-100 Weldments (base with option)

1. Description of Services/introduction

Naval Surface Warfare Center, Carderock Division (NSWCCD) Codes 61, 62, and 65 are investigating the effects of ultrasonic impact peening treatment (UIT) and hammer peening on the fatigue performance of HY-100 weldments. To support this investigation, the NSWCCD needs to procure procedure development, UIT and hammer peening services. This purchase, and subsequent testing, supports NAVSEA PMS450 Task No. 450-T-0055 Welding Fatigue Improvement.

2. Objective

The objective of this purchase is to procure procedure development, UIT and hammer peening services from a single contractor. The contractor shall develop the treatment procedure and provide to the Government for approval. The Government will provide Tee and Butt weldments to the contractor for UIT and hammer peening treatments. The weldments will be provided in batches, described in Section 2.3. For base contract Initial Batch Treatments, the contractor shall perform both UIT and hammer peening, per Section 2.3.1. The Government will purchase Second Batch Treatment services, either UIT or hammer peening, as an exercised contract option, per Section 2.3.2. The full list of services required and requirements for each service are provided below.

2.1 Contractor Responsibilities

- The contractor shall only perform work as directed in the official purchase order (PO) issued by contracting agency (NSWCCD). This technical statement of work does not in itself constitute a binding contract.
- The contractor shall adhere to commercial best practices for maintaining specimen identification throughout the machining and packaging processes. Specimens will be identified and labeled upon delivery to the contractor. The contractor shall reference the provided IDs to track which treatments are applied to which specimens.
- The contractor shall handle all Government furnished material with care so that no damage, scratches, gouges, or dents, are inflicted on the material.
- The contractor shall be responsible for initiating the UIT and hammer peening within 5 business days of the date that the samples are received or the treatment plan is approved by NSWCCD, whichever date comes second.
- The contractor shall complete all UIT and hammer peening treatments, for each batch provided by The Government, within 2 months of receipt of material.
- The contractor shall ship treated weldments back to NSWCCD. All shipped material shall be packaged to preserve the integrity and maintain the identity of the specimens.
- The contractor shall be responsible for submitting invoices to the Government within two months from the time of Government review and acceptance of all treated weldments.

2.2 Hammer Peening and UIT Treatment Requirements

- The contractor shall document and develop a treatment plan for the weldments based on industry best practices and provide it to the NSWCCD for review and approval.
 - The treatment plan shall cover hammer peening and UIT treatments for the provided butt and tee weldments.
- The contractor shall apply both types of weld treatments with robotic tools.
- The contractor's UIT equipment shall be powered by a magnetostrictive transducer.
- The contractor's UIT equipment shall be able to perform at a 100% duty cycle.
- The contractor shall use pneumatic equipment for hammer peening treatments.
- For each specimen, the contractor will apply the treatment to every weld toe. There are 2 weld toes on each tee joint, and 4 weld toes on each butt joint.

2.3 Treatment Quantities

2.3.1 (BASE CONTRACT) Initial Treatment Batch

Treatment Type	Tee Weldments	Butt Weldments
Hammer Peening	8	8
UIT Peening	22	16

2.3.2 (OPTION) Second Treatment Batch

The weldments provided in the second treatment batch will be treated with either hammer peening or UIT peening. The treatment method is dependent on Government testing that will take place after the first treatment batch is returned. The Government will confirm the treatment method upon exercise of option and prior to shipping the second treatment batch.

Treatment Type	Tee Weldments	Butt Weldments
Hammer or UIT peening	8	8

2.4 Return of Materials

The contractor shall ship all material back to the Government at the performer's expense within 10 business days of the treatment completion date of all Government provided weldments. Completed weldments shall be shipped to:

Attn: TBD At Award
Naval Surface Warfare Center, Carderock
Code 612, Building 60, Room 172
9500 MacArthur Boulevard
West Bethesda, MD 20817

3. Special Qualifications

3.1 The contractor shall be a US based company.

4. Specimen Tracking and Labeling

NSWCCD will label all specimens prior to shipping them to the contractor site for treatment. The contractor shall not remove any existing labels from the weldments, and the contractor shall reference the provided ID numbers whenever necessary.

5. Government Furnished Material

The Government will ship the weldments to the contractor in two batches, quantities are defined in section 2.3. The approximate size of each weldment is 35 inches x 5 inches x 1.5 inches. The weld length is approximately 5 inches long. The weldments are made of HY-100 material welded with MIL-100S filler material. The Government may provide scrap material for the contractor to use in setup.

5.1 Butt Weld Specimen Geometry

All provided butt weld specimens will be the same size.

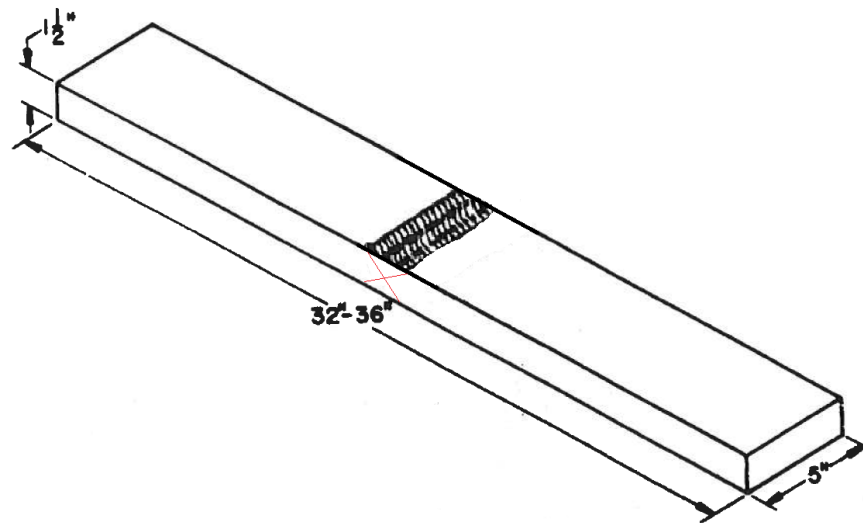


Figure 1. Butt Joint Geometry

5.2 Tee Weld Specimen Geometry

All provided tee weld specimens will be the same size.

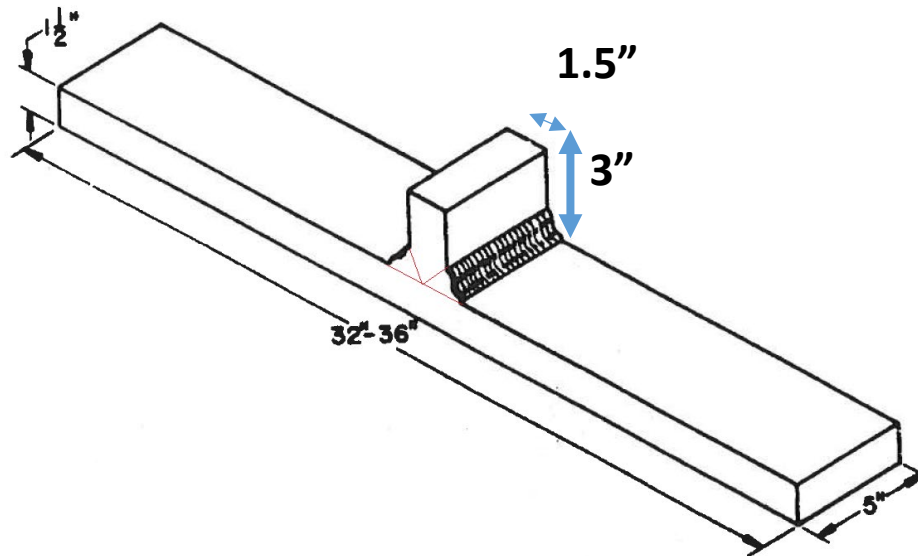


Figure 2. Tee Joint Geometry

6. Deliverables

6.1 CDRL A001, Treatment Plan – Hammer and Ultrasonic Peening

The contractor shall provide a treatment plan that includes the hammer peening and ultrasonic impact treatment procedures that will be applied to the described weldments. Requirements are defined in Section 2.2. The plan shall be signed by the cognizant senior engineer overseeing this project. The contractor shall electronically provide the treatment plan to the Government TPOC two weeks following award of contract. Within three weeks of receipt of procedures, the TPOC will approve the procedures as submitted, or request the contractor provide revisions within timeframe determined by the TPOC. If procedures need revisions the contractor shall provide the procedures within three weeks and resubmit to the TPOC for review.

7. Period and Place of Performance

7.1 The period of performance for the base contract is 12 months from the date of contract award. The period of performance for the option is one year from time of option award.

7.2 Work associated with this PWS will occur at the contractor site.

8. Security Requirements

All work on this contract shall not exceed the classification UNCLASSIFIED.

9. Government Technical Point of Contact

TBD At Award